

Title: Resultant Force

Starter

1. What is the unit for forces?
2. What does it mean if a force is balanced?
3. What does it mean if a force is unbalanced?

forces

balanced forces

equal strength

opposite directions

object is at rest

equal forces

act on
objects
as push or pull

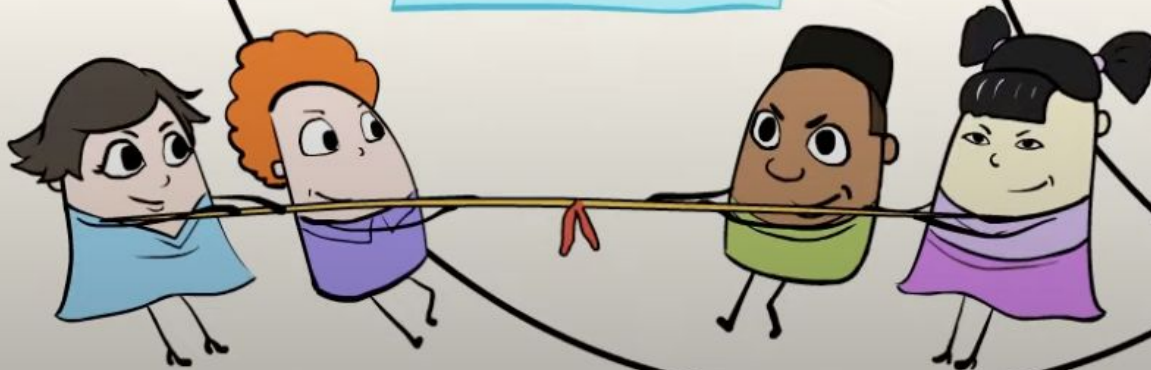
unbalanced forces

cause the object to
accelerate or decelerate

cause a change in the
direction of an object

make the object
start or stop moving

not equal forces

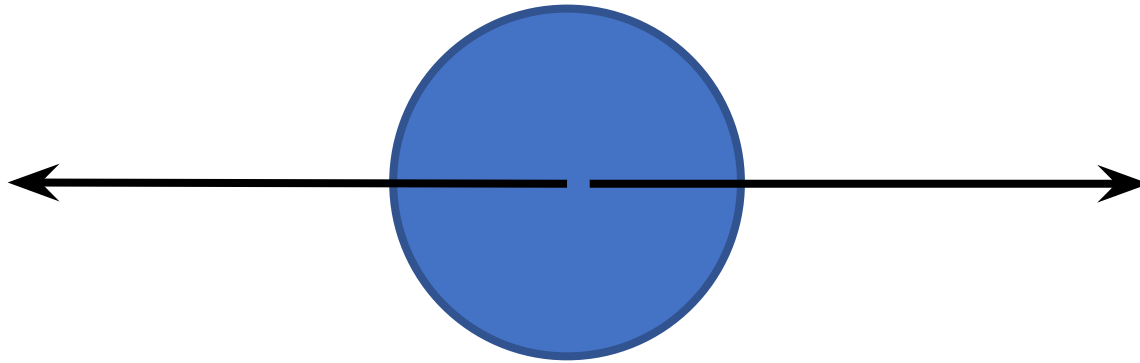


Success criteria

Lesson Title	Working towards	Working at	Working beyond
Recap: Forces & Motion (Y7)	I can identify different forces acting on an object.	I can describe what balanced and unbalanced forces are.	I can explain how unbalanced forces change an object's motion using force diagrams.
Speed	I can identify the correct units for speed, distance, and time.	I can use the speed = distance $\div$ time formula to calculate speed, distance or time.	I can solve speed questions using different units and rearranged formulas.
Describing Movement	I can read values from a distance-time graph.	I can describe motion based on the shape of the graph.	I can draw and explain distance-time graphs from written data or scenarios.
Turning Forces	I can give examples of things that turn when a force is applied.	I can use the moment formula to calculate turning forces.	I can explain how balanced and unbalanced moments affect turning.
Pressure between solids	I can state that heavier or larger objects create more pressure.	I can calculate pressure using the correct formula and units.	I can compare pressure in different situations using scientific reasoning.
Pressure in liquids and gases	I can describe how pressure changes with depth in liquids and altitude in gases.	I can explain what causes pressure in liquids and gases using particles.	I can calculate liquid pressure using the h ρ g formula and explain examples.
Particles on the move	I can define diffusion.	I can describe how temperature and concentration affect diffusion.	I can explain diffusion using the particle model and apply it to real examples.

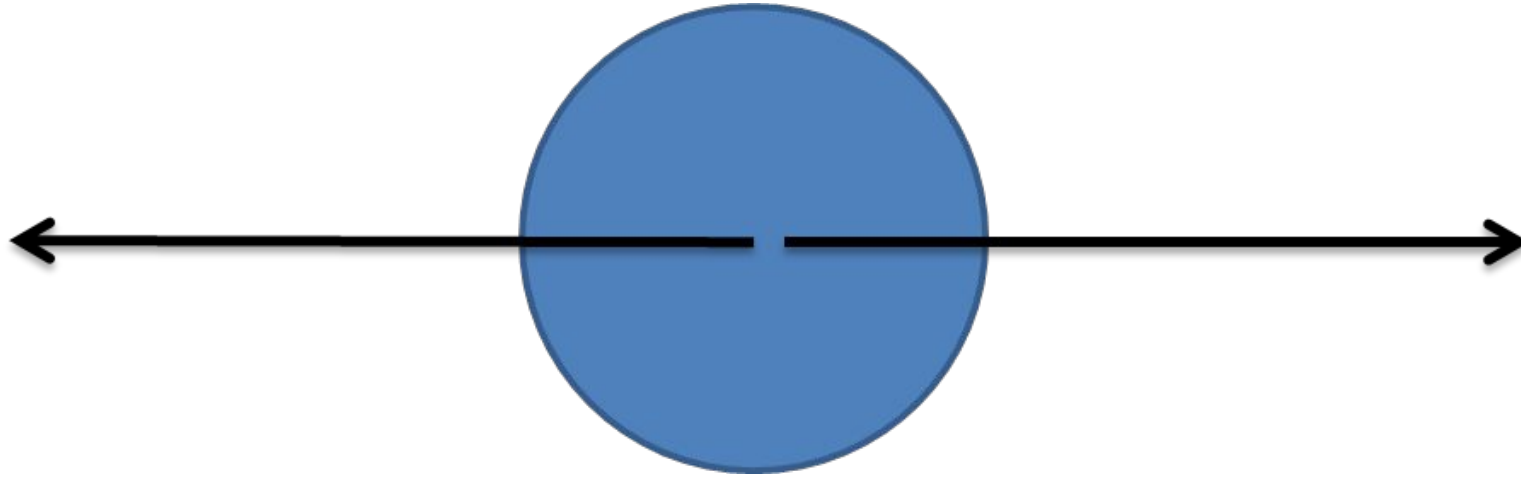
Force Diagrams

- The arrow points in the direction the force is trying to move the object
- The size of the arrow represents the size of the force



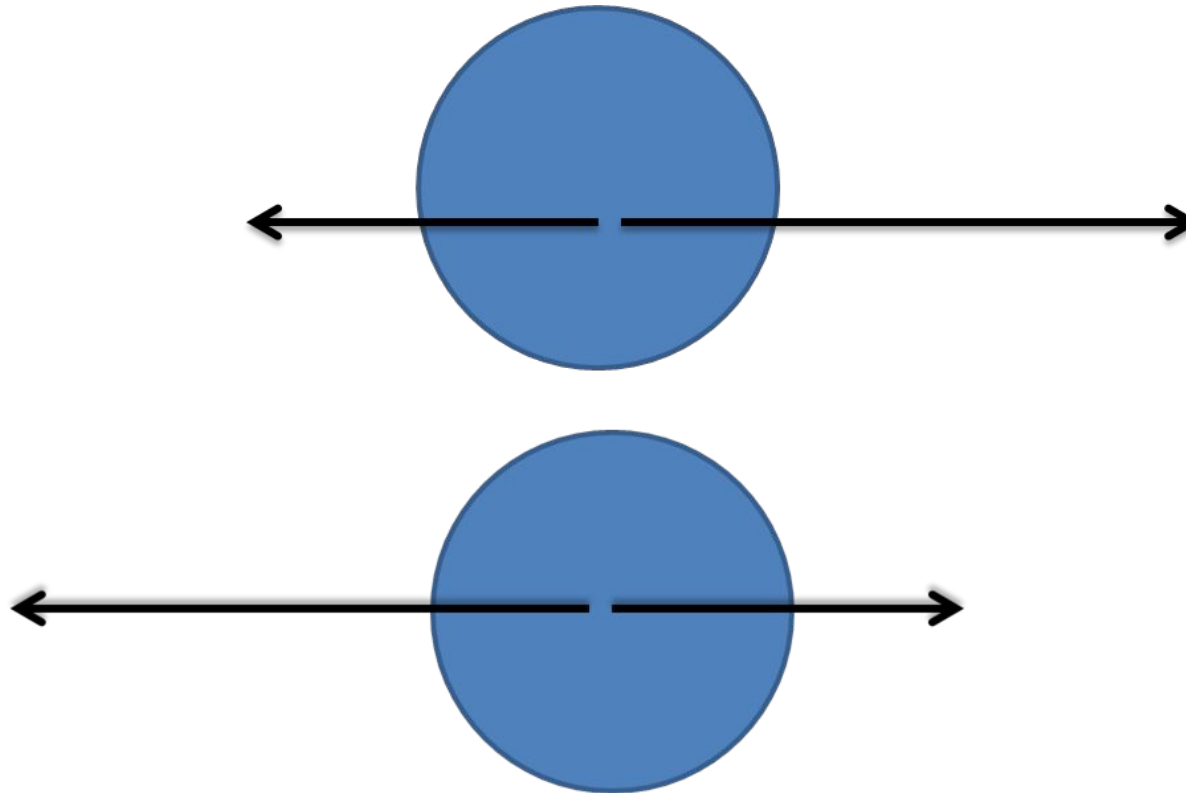
Balanced forces cause the object's
speed to _____

(change/ stay the same)



Unbalanced forces cause the object's speed to _____

(change/ stay the same).



Unbalanced forces

Unbalanced forces cause the objects speed (or direction) to change.

If the object is stationary?

- The object starts moving.



Unbalanced forces

Unbalanced forces cause the objects speed (or direction) to change.

What if the object is already moving?

- The object speeds up or slows down



Balanced forces

Unbalanced forces cause the objects speed (or direction) to change.

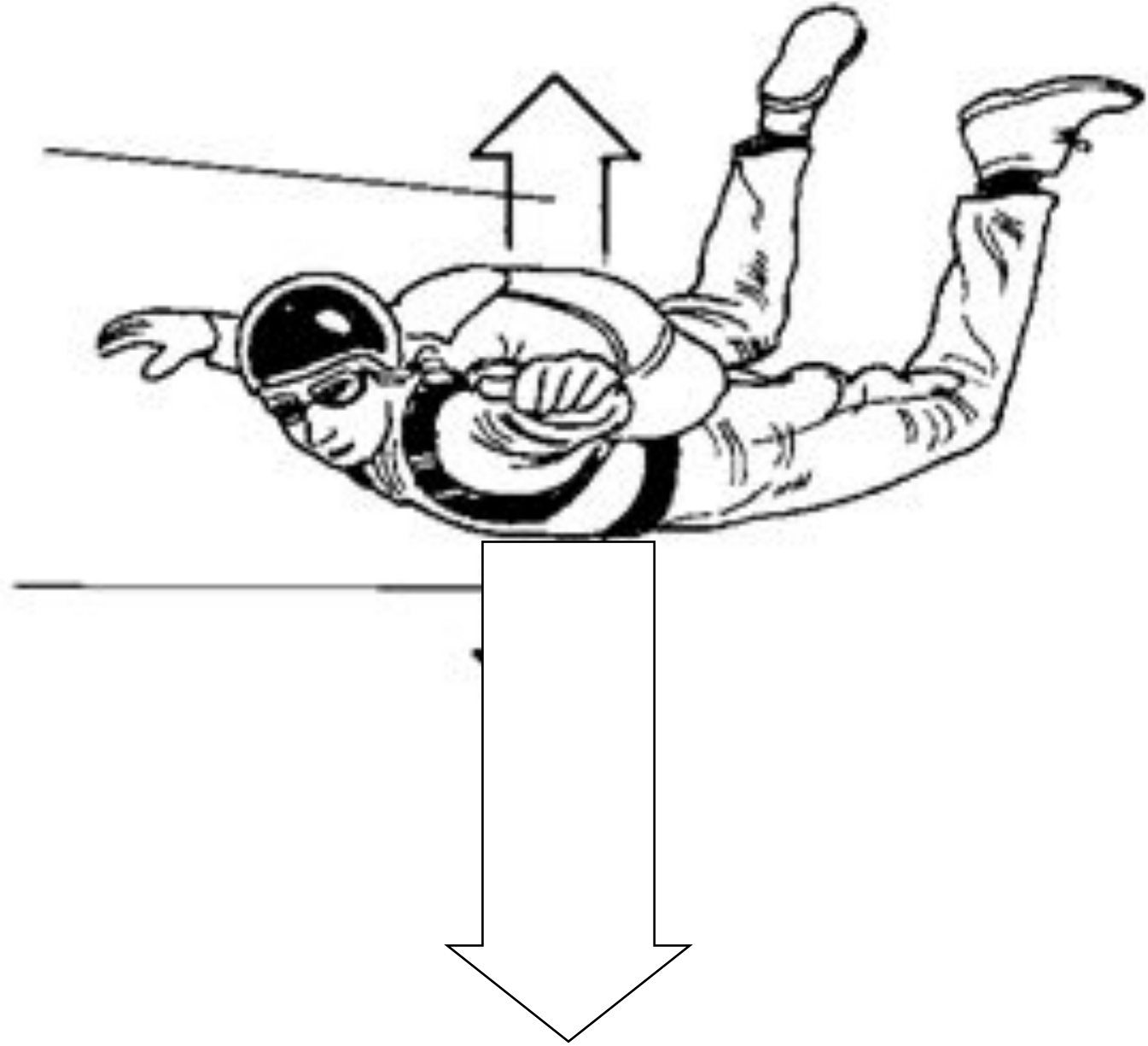
What if the forces are balanced?

The car is either stationary or moving at a constant speed



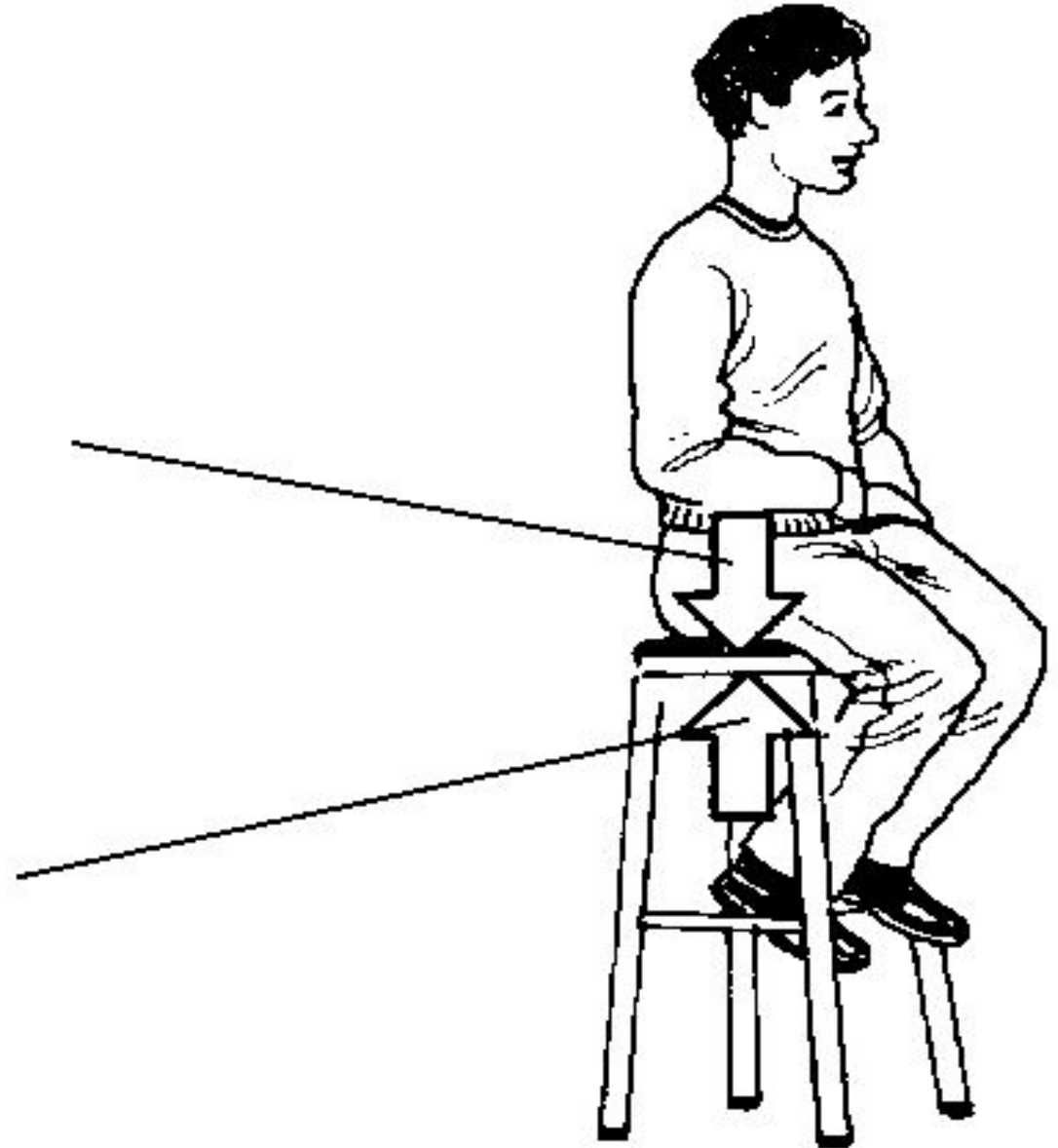
Are the forces balanced or unbalanced in this picture?

Can you describe the motion?



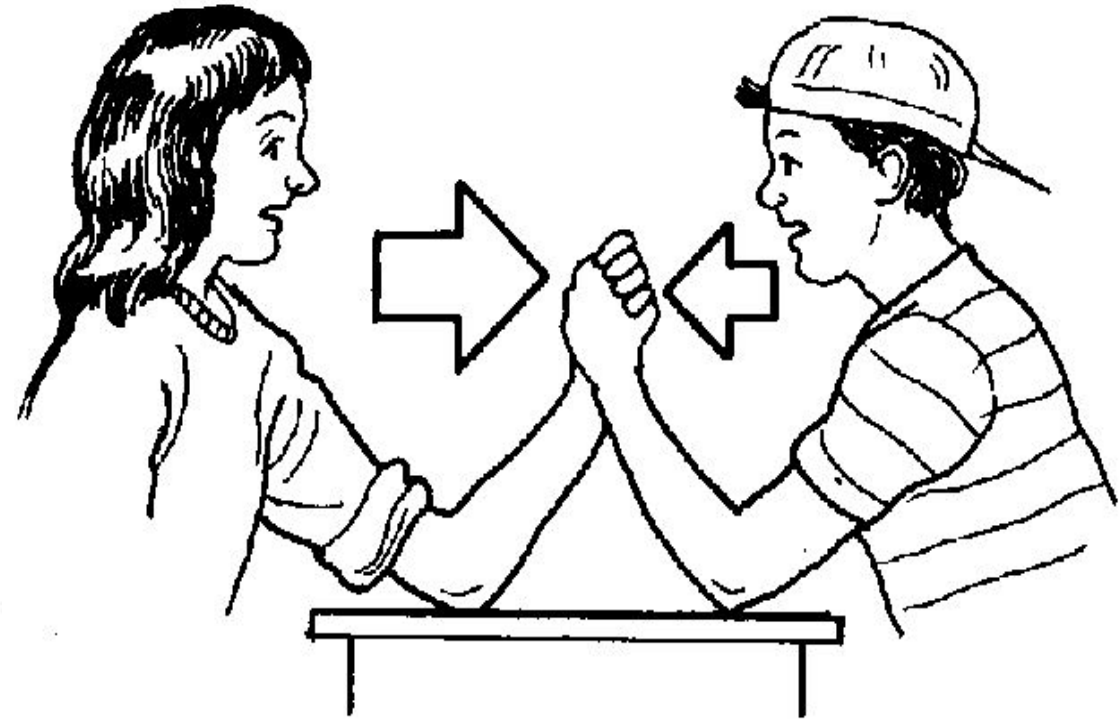
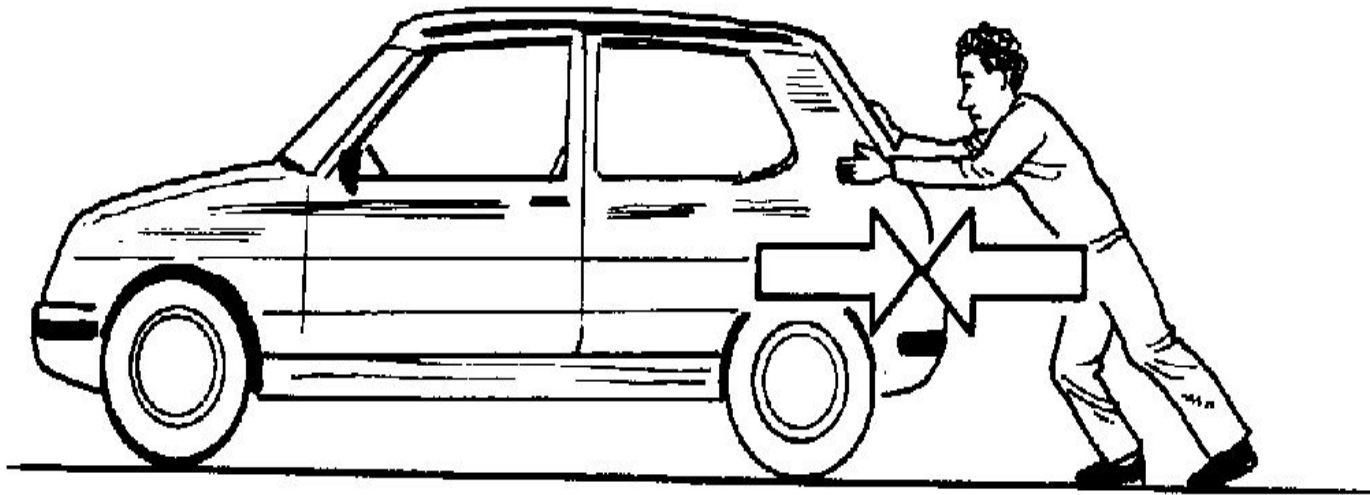
Are the forces balanced or unbalanced in this picture?

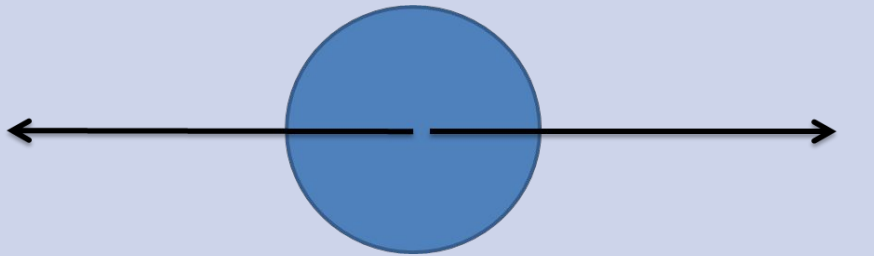
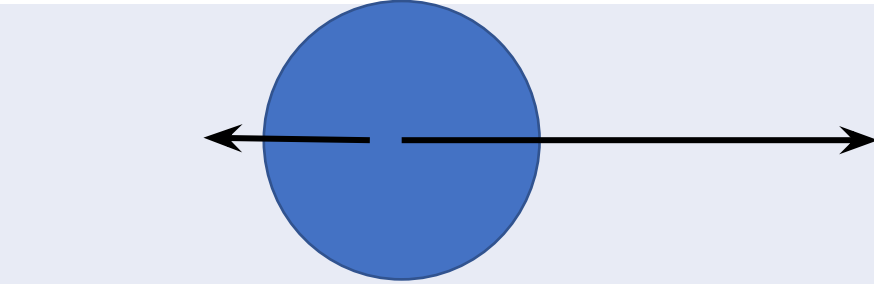
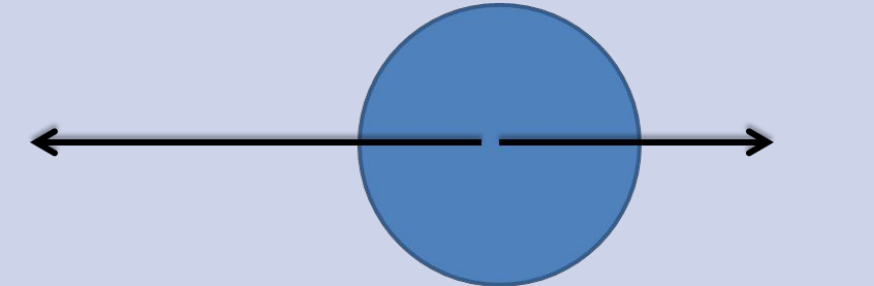
Can you describe the motion?



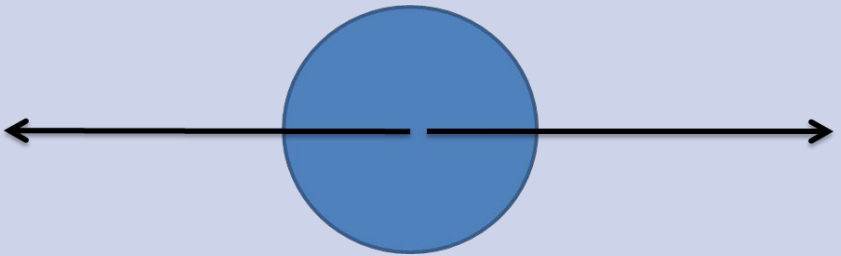
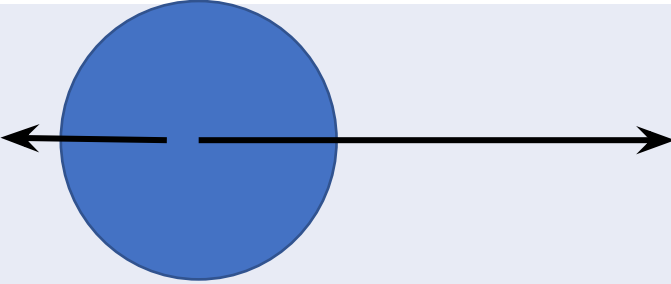
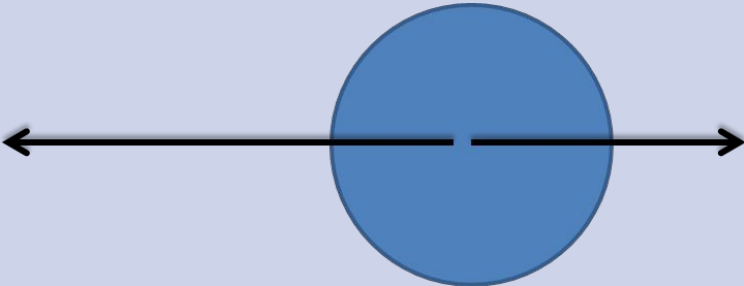
Are the forces balanced or unbalanced in these pictures?

Can you describe the motion



Force Diagram	What happens to the ball?	Balanced or Un-balanced?
		
		
		

10:00

Force Diagram	What happens to the ball?	Balanced or Un-balanced?
	Constant speed // Not moving, stationary	balanced
	Moving to the right	unbalanced
	Moving to the left	unbalanced

Newton's 1st law of motion

An object will not speed up or slow down unless an external force acts on it

1. How much an object speeds up or slows down depends upon the force applied, and the mass of the object
2. When two objects exert a force on each other, the effects are equal and opposite



Resultant force

- The resultant force is the **sum** (total) of all forces acting on an object. This net force causes the change in speed, shape or direction.

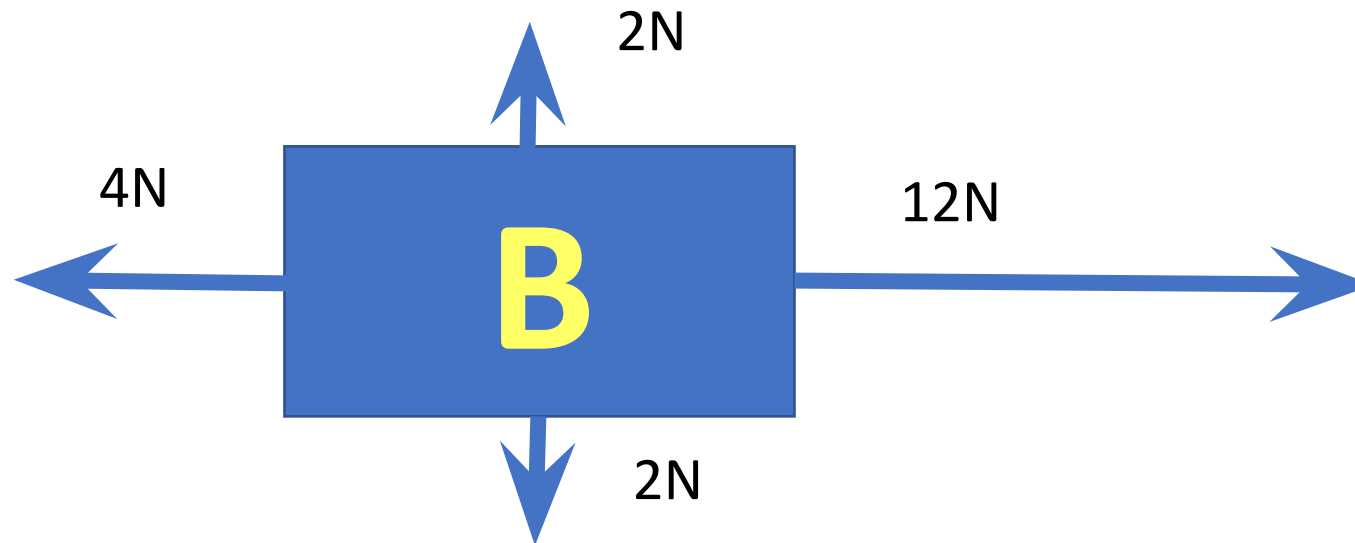
Resultant Forces (RF)

- The resultant force is the sum of all forces acting on an object. This net force causes the change in speed, shape or direction.



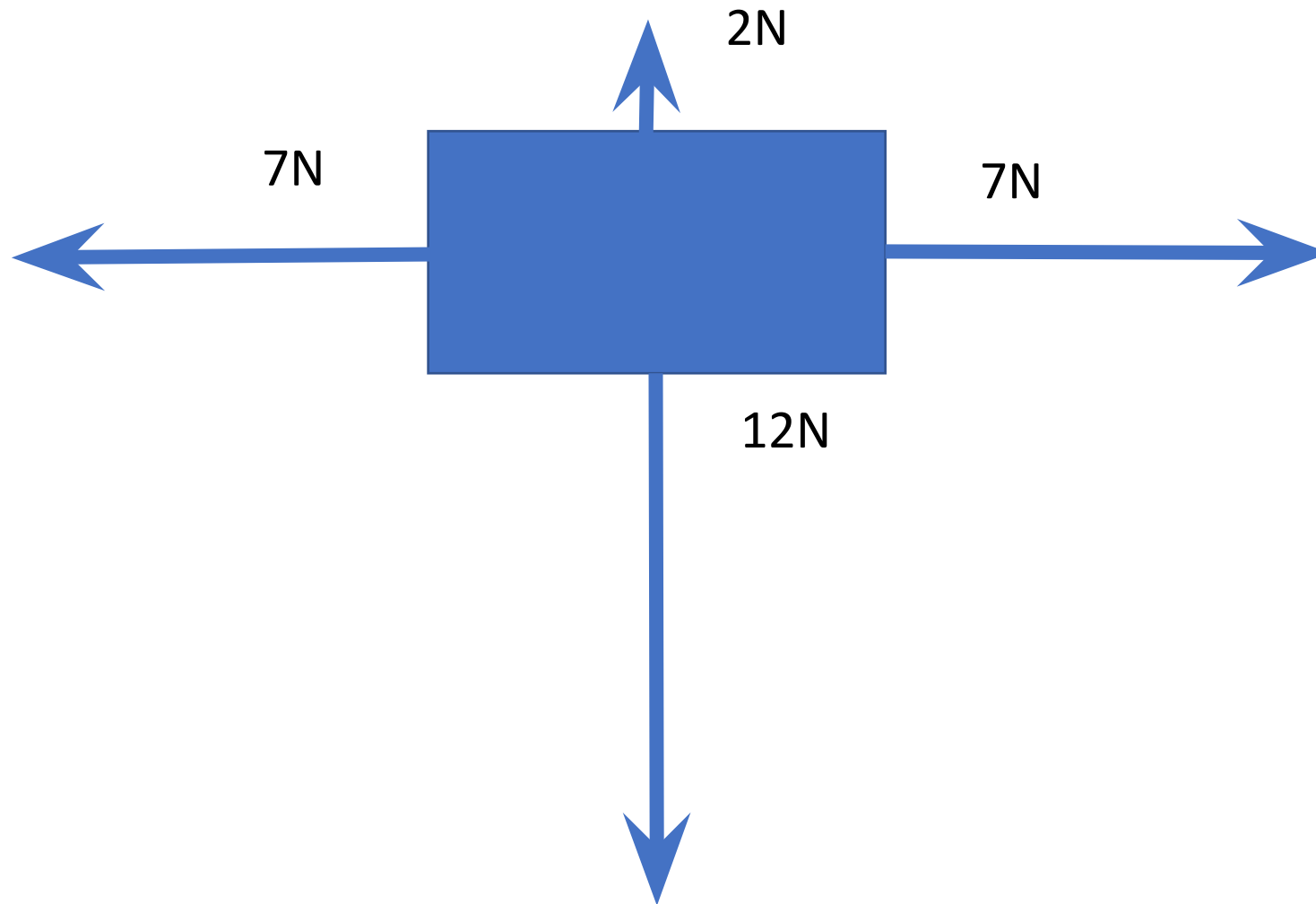
- Balanced or unbalanced?

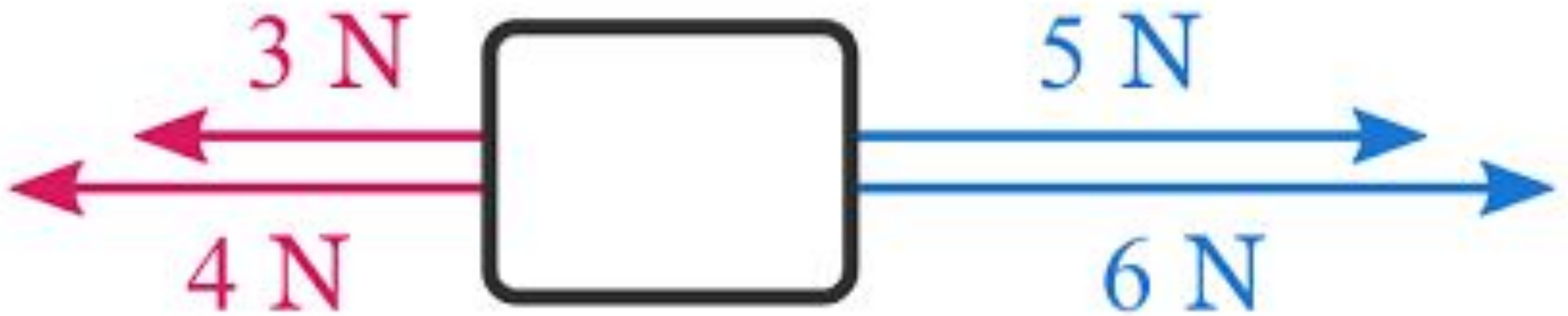
Resultant Forces (RF)



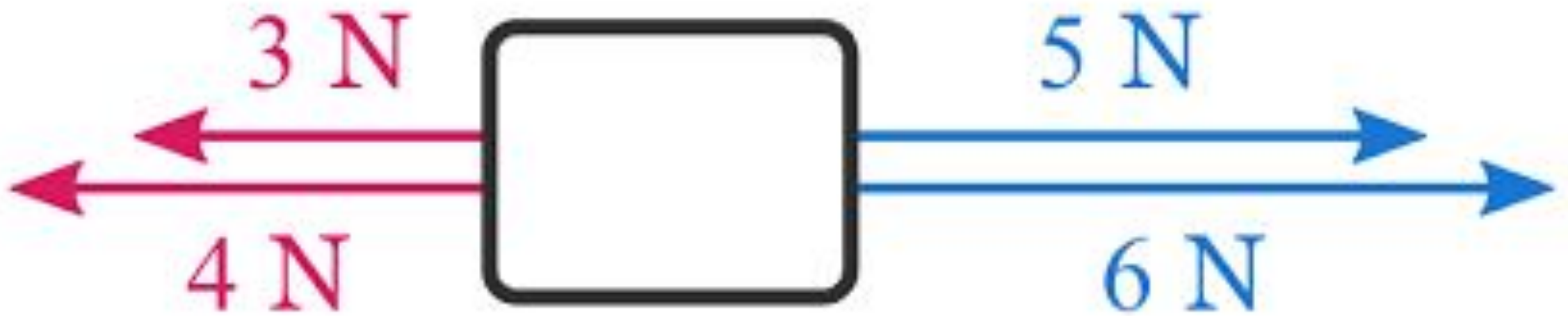
• Balanced or unbalanced?

Resultant Forces (RF)

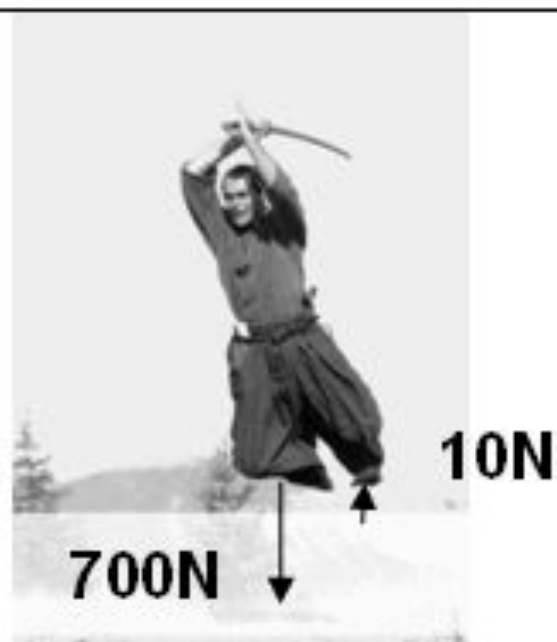




Calculate the resultant force above



11N - 7N = 4N to the right



This man is jumping through the air, are the forces on him balanced or unbalanced?

.....
Which forces are acting on the man?

..... and
What happened to the man after the photo was taken?

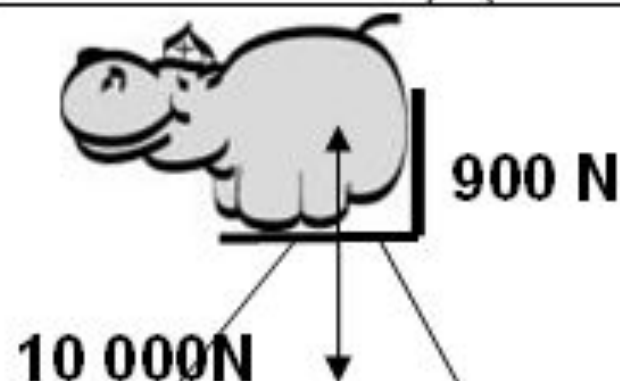
.....
Balanced or unbalanced?

.....
Which direction will the toy move?

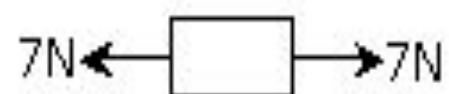
.....
Balanced or unbalanced?

.....
What will happen to the chair? What will happen to the hippo?

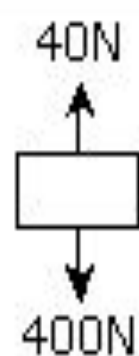
.....



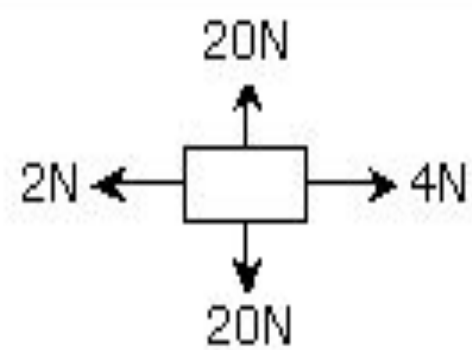
a.



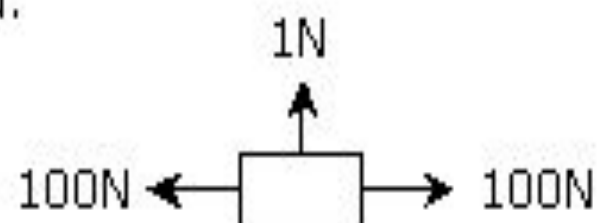
b.



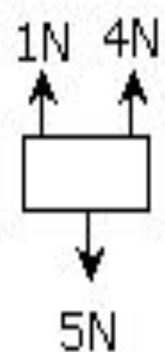
c.



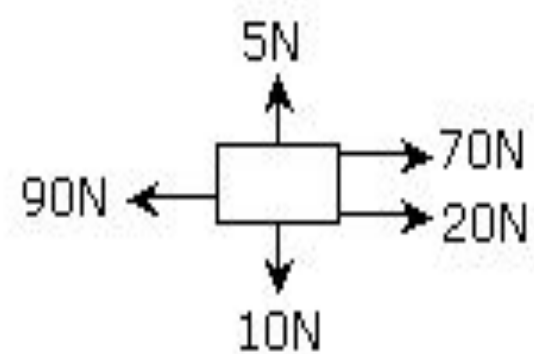
d.



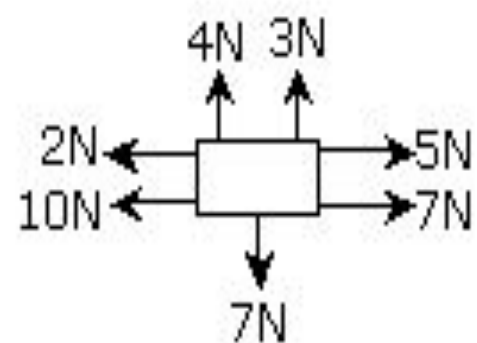
e.



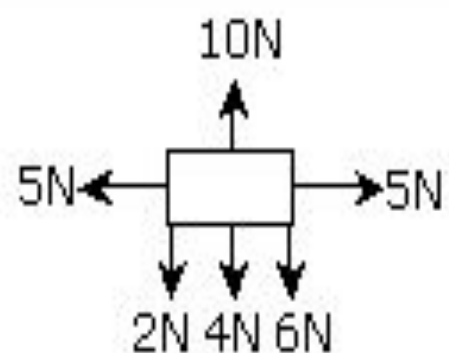
f.



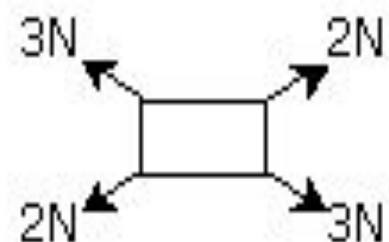
g.



h.



i.





This man is jumping through the air, are the forces on him balanced or unbalanced?

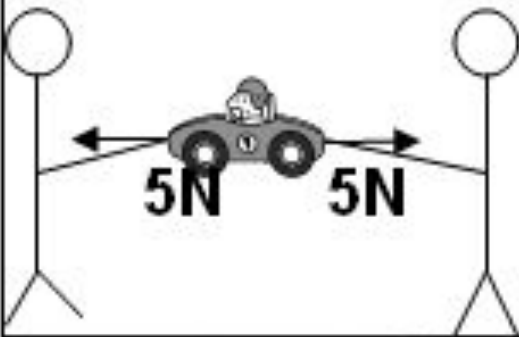
unbalanced

Which forces are acting on the man?

Gravitational force and **Air resistance**

What happened to the man after the photo was taken?

He falls downwards towards the ground because weight is much greater than the upward force.

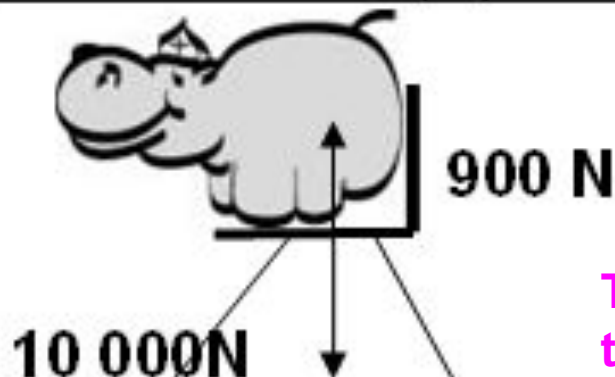


Balanced or unbalanced?

balanced

Which direction will the toy move?

It will not move — it stays still (forces are equal and opposite).

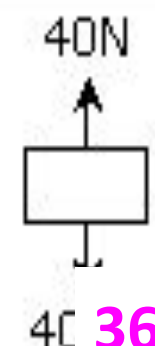
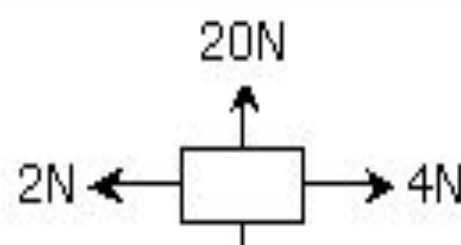
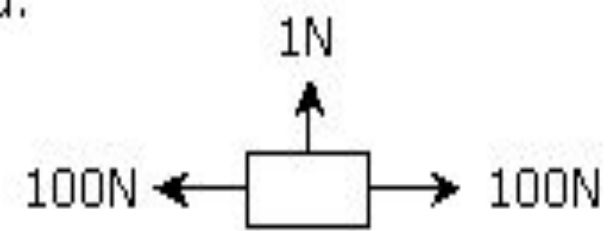
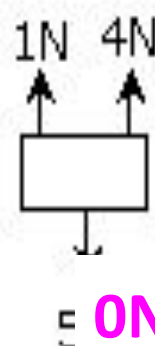
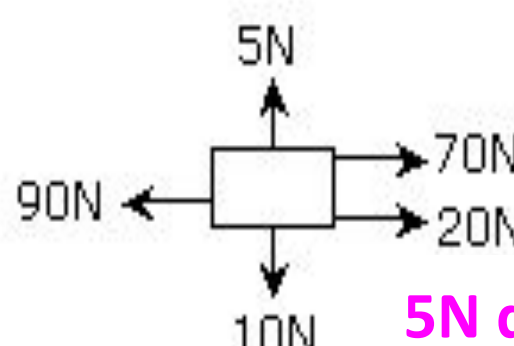
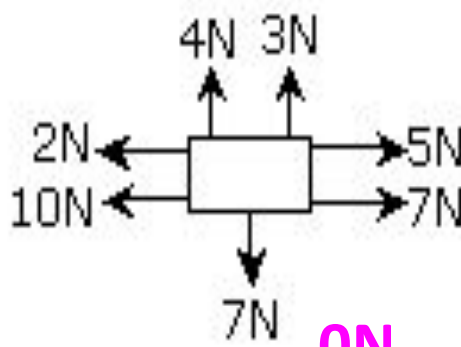
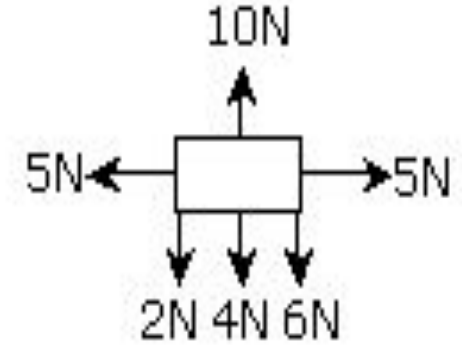
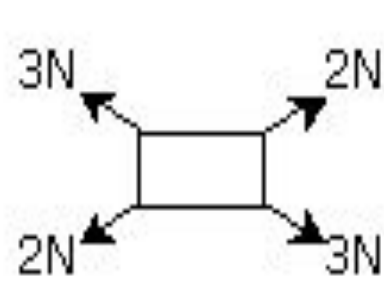


Balanced or unbalanced?

unbalanced

What will happen to the chair? What will happen to the hippo?

The chair will break or collapse because the hippo's weight is much larger than the upward support force. The hippo will fall downwards to the ground.

a.  0N	b.  360N down	c.  2N right
d.  1N up	e.  0N	f.  5N down
g.  0N	h.  2N down	i.  0N

Experiment 2: Using the force-meters

SEE PAGE 74-75 of the purple learner's book for instructions.

Think like a scientist

Measuring balanced and unbalanced forces

You will investigate the effects of balanced and unbalanced forces.

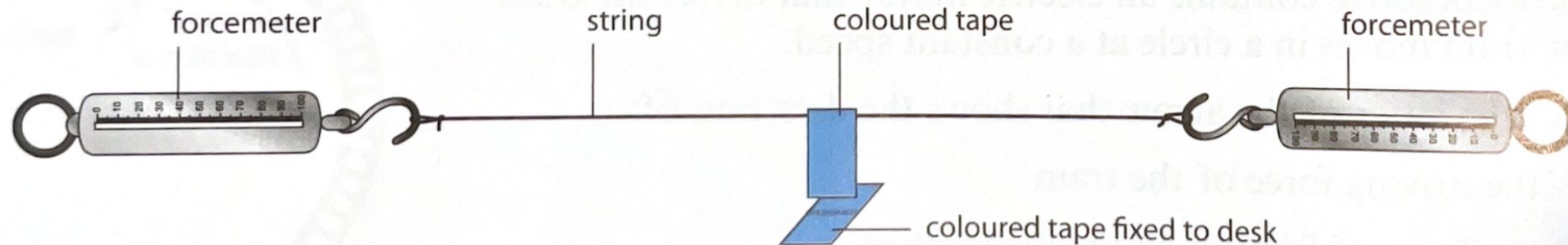
Work in pairs.

Be Gentle! Do not overstretch the Spring!

You will need:

- two forcemeters
- piece of string
- coloured tape
- scissors

Set up the equipment as shown in the diagram.



Method

- 1 Each person in the pair holds one forcemeter (also known as a newton meter). The string should be tight.
The coloured tape on the string should be lined up with the coloured tape on the desk.
- 2 Each person pulls with an equal force, for example, 4 N.

**Be Gentle! Do not
overstretch the Spring!**

Think, Pair and Answer the following questions!

- 1 What directions must you both pull to keep the pieces of coloured tape lined up?
- 2 Both people increase the pulling force to, for example, 8 N.
Explain why the string does not move, even when the force is increased.
- 3 One person decreases their force by 1 N. If the force was 8 N, then decrease the force to 7 N.
 - a Describe what happens to the string.
 - b Use a force diagram to explain what happens to the string.
- 4 Now make the difference between the forces larger so that, for example, the difference is now 2 N or 3 N.
How does the difference between the sizes of the forces affect the movement of the string?
- 5 Explain why the two forcemeters do not have to be the same.
- 6 This investigation is an analogy of a tug of war. That means the investigation can be compared with a tug of war.
State what is represented by the forcemeters in this analogy.

Check your answers!

1. In **opposite directions** with equal force.
2. Because the forces are **balanced** — equal size, opposite direction.
3. The string will move towards the person pulling harder (the 8 N side).
4. Draw two arrows in opposite directions:
 - One arrow = 8 N (longer)
 - Other arrow = 7 N (shorter)

The arrows are not equal, so the resultant force is **1 N towards the stronger side.**
5. The larger the difference, the faster the string moves towards the stronger side
6. Any forcemeter can be used because both measure force in newtons.
7. The forcemeters represent the **people pulling on each side** in a tug of war.

Workbook page 38-41



Unit 3 Forces and energy

Topic 3.1 Forces and motion

Exercise 3.1A Balanced forces

1

Statement	Needed for forces to be balanced?
Two forces must be the same size.	✓
Two forces must be different sizes.	
Two forces must be in the same direction.	
Two forces must be in opposite directions.	✓

- 2
- a false
 - b true
 - c false

Check your answers!

- 3
- Forward arrow and backward arrow of approximately the same length; forward arrow labelled driving force; backward arrow labelled friction. Weight and contact force need not be shown as they were not asked for.

Exercise 3.1B Unbalanced forces

Check your answers!

- 1 Ticks by 'a boat will slow down', and 'a football will change direction'.
- 2
 - a contact force and weight
 - b driving force and friction
 - c start to move forward/get faster/speed will increase

Exercise 3.1C Changing direction

Check your answers!

- 1
 - a Bird is flying at a constant speed and at a constant height.
 - b Force **A** increases (credit can be given to force **C** decreasing, although it can be pointed out that this is less likely).
 - c Force **D** increases; force **B** decreases.
- 2 arrow pointing to the right of the page
- 3
 - a Arrow on the string pointing toward the pole labelled F.
 - b Arrow at a tangent to the circle from the ball in the same direction as the rotation of the ball labelled D.

Plenary



Which car will:

- 1) Travel at steady speed
- 2) Accelerate (get faster)
- 3) Slow down (decelerate)